should be underlined that, because with vaccination, the emergence of frequent O groups not included in the vaccine is likely, as happened with the pneumococcal conjugate vaccines [8].

These findings highlight the importance of performing regular molecular epidemiology studies from different regions to predict vaccine coverage. Adding the O17 serogroup to the vaccine, if possible, appears to be an interesting strategy.

Notes

Ethical approval and consent to participate. The COLIBAFI study was approved by the French Comité de Protection des Personnes of Hôpital Saint-Louis, Paris, France (approval #2004–06, June 2004). The SEPTICOLI study was approved by the French Comité de Protection des Personnes Ile de France no. IV (IRB 00003835, March 2016).

Because of their noninterventional nature, only an oral consent from patients was requested under French law.

The whole-genome sequences of the 912 strains studied have been deposited under the Bioprojects PRJEB35745 and PRJEB39260.

Financial support. This work was partially funded by a Translational Research Grant from the Agence Nationale de la Recherche (ANR), Ministry of Higher Education and Research, France, 2015 (grant no. ANR-15-CE-17-0019-01), a grant from INSERM (INSERM grant RBM-03-58), and the Projet Hospitalier de Recherche Clinique (Assistance Publique-Hôpitaux de Paris [AP-HP] grant AOR 04 053). E. D. was partially supported by the "Fondation pour la Recherche Médicale" (Equipe FRM 2016, grant number DEQ20161136698). G. R. was supported by a "Poste d'accueil" funded by the AP-HP and a Commissariat à l'Energie Atomique et aux Energies Alternatives (CEA) personal grant for his PhD.

Potential conflicts of interest. The authors: No reported conflicts of interest. All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

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Clinical Infectious Diseases® 2022;74(2):364–6
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DOI: 10.1093/cid/ciab458

Neutralization of Variant Under Investigation B.1.617.1 With Sera of BBV152 Vaccinees

To the Editor—The emergence of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) variants in places where the virus is uncontained poses a global threat from the perspective of public health and vaccine efficacy. Peng et al [1] recently reported on the increased transmissibility with the newly emerged Variant of Concern (VOC) (20C/S:452R and 20C/S:452R) with the L452R mutation in San Francisco. We report the immunological characteristics of a Variant under Investigation (VUI) B.1.617.1, playing a critical role in the current surge of coronavirus disease 2019 (COVID-19) in the western state of Maharashtra, India.

Several SARS-CoV-2 variants—B.1.1.7, B.1.351, and B.1.1.28.1—have been reported in India during 2021 [2, 3]. We had sequenced 146 nasopharyngeal/oropharyngeal swabs of COVID-19 cases [4]. Among these, 15 sequences had a combination of L452R and E484Q mutations, which raised concern as both are found in the receptor-binding domain (RBD) of the spike protein. However, the combined effect of these mutations is still unknown.

A total of 23 nonsynonymous changes were observed in the sequences, out of which 7 conserved nonsynonymous changes were found at the spike protein (G142D, E154K, L452R, E484Q, D614G,

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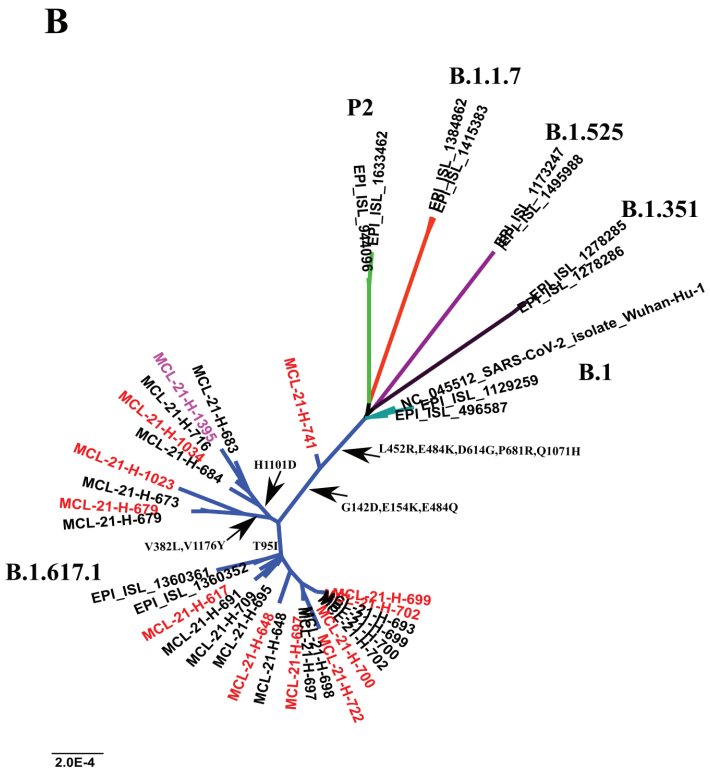
P681R, Q1071H) along with other conserved mutations (Figure 1A). These sequences were classified as VUI B.1.617.1, a sub-lineage of B.1.617. Phylogenetic analysis of B.1.617.1 sub-lineage sequences led to distinct clusters. The amino acid mutation analysis of the spike protein sequences retrieved from the study revealed them to have independent common

A

Reference Position	CDS/Gene	Reference	Allele	Amino acid change
241	5' UTR: 5' UTR	C	T	
3037	CDS: ORF1ab, former nsp1	C	T	
3457		C	T	
4965		C	T	Thr1567Ile
11201		A	G	Thr3646Ala
14408	CDS: ORF1ab, nsp12	C	T	Pro4715Leu
17523	CDS: ORF1ab, nsp13	G	T	Met5753Ile
20396	CDS: ORF1ab,nsp15-A1 and nsp15B	A	G	Lys6711Arg
21895	CDS:Spike	T	C	
21987		G	A	Gly142Asp
22022		G	A	Glu154Lys
22917		T	G	Leu452Arg
23012		G	C	Glu484Gln
23403		A	G	Asp614Gly
23604		C	G	Pro681Arg
24775		A	T	Gln1071His
25469	CDS: ORF3a	C	T	Ser26Leu
27638	CDS: ORF7a	T	C	Val82Ala
28881	CDS: N	G	T	Arg203Met
29402		G	T	Asp377Tyr
29742	Stem loop: basepair exception	G	T	



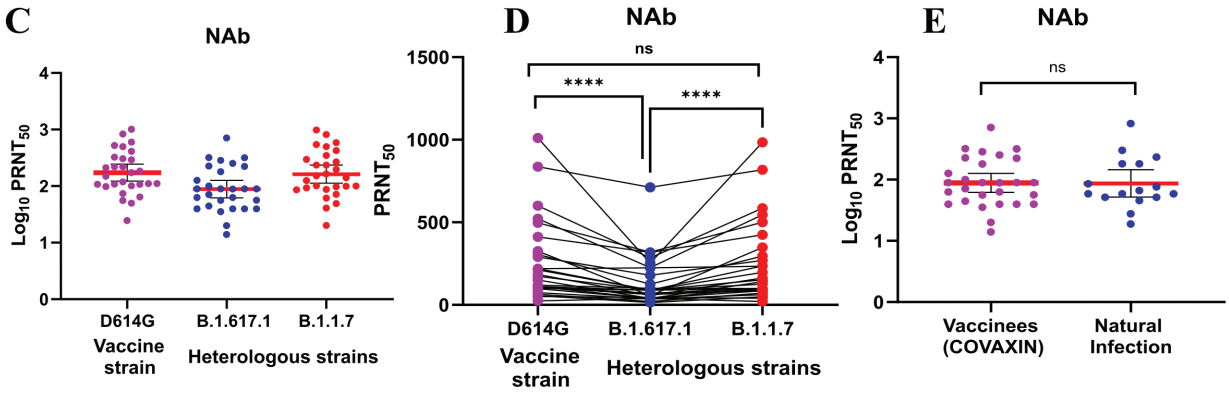


Figure 1. Characteristics and neutralization of the VUI B.1.617.1 variant. *A*, The common nucleotide changes observed in the majority of the isolates and clinical sequences from B.1.617.1 sub-lineage. *B*, A neighbor-joining tree was generated using a Tamura 3-parameter model with gamma distribution and a bootstrap replication of 1000 cycles. Isolates are marked in red and sequences from foreign travelers marked in pink. The representative sequences from other clades are represented as B.1.1.7 (red), B.1.617.1 (blue), B.1.351 (black), B.1.525 (purple) Brazil P2 (light green), and B1 (light blue). Individual spike mutations specific to the clusters are marked using the arrows. One of the sequences (MCL-21-H-741) had an E484K mutation, which led to its distinct clustering, in B.1.617.1 lineage (blue) that belonged to a traveler who had returned from United Arab Emirates to India. *C*, Scatterplot depicting the neutralizing response of the individual sera (n = 28) vaccinated with BBV152 (Covaxin, Bharat Biotech International Ltd., India) collected during a phase II clinical trial for the prototype B1 (D614G) (pink), B.1.1.7 (red), and B.1.617.1 (blue). The red solid line indicates the geometric mean titer and the error bar depicts the 95% confidence interval. *D*, Neutralization of the matched-pair samples compared with prototype D614G (pink), B.1.1.7 (red), and B.1.617.1 (blue). Neutralization reduction by a factor of 1.95 and 1.8 was observed against the B.1.617.1 variant for B1 (D614G) and B.1.1.7 variants, respectively. A reduction factor of 1.06 was observed between the B1 (D614G) and B.1.1.7 variant. A two-tailed pairwise comparison was performed using the Wilcoxon matched-pairs signed-rank test with a *P* value of .05. ****$P < .001$, ns = nonsignificant P value. *E*, Neutralization of the COVID-19 -recovered cases' sera (n = 16) of B.1.1.7 (n = 2), B.1.351 (n = 1), B.1.1.28.2 (n = 2)–infected individuals PRNT50 values against B.1.617.1 variant was compared with vaccine recipient serum samples. A two-tailed pairwise comparison was performed using the Mann-Whitney test with a *P* value of .05. Abbreviations: COVID-19, coronavirus disease 2019; PRNT, plaque reduction neutralization test; UTR, untranslated region CDS: Coding sequence; NAb: Neutralizing antibody.

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mutations: first, T95I; second, H1101D; and third, V382L and V1175Y (Figure 1B). So far, 21 countries have reported the presence of B.1.617 variants [5]. Virus isolation was attempted from 15 specimens using Vero-CCL-81 cells [6]. Twelve specimens displayed cytopathic effects on the fourth post-infection day, which were further passaged and titered for performing a plaque reduction neutralization test (PRNT) [7]. These isolates were obtained from clinical specimens of asymptomatic individuals (age range: 14–55 years) and cases with a low-grade fever, cough, and sore throat (age range: 26–77 years).

The neutralization efficacy of the VUI B.1.617.1 variant was compared with B1 (D614G) and B.1.1.7 variants using sera of 28 BBV152 vaccinated individuals, collected during the phase II clinical trial [8]. The D614G versus B.1.617 Geometric mean titer (GMT) ratio was 1.95 (95% confidence interval [CI]: 1.60–2.38; $P < .0001$). Similarly, the GMT ratio comparison of B.1.1.7 was significantly higher than the GMT for B.1.617.1 (GMT ratio: 1.84; 95% CI: 1.50–2.27; $P < .0001$) (Figure 1C and 1D). The comparison of D614G and B.1.1.7 showed equivalent responses, with a GMT ratio of 1.06 and a 95% CI of 1.02–1.10.

Sera samples collected from individuals recovered from COVID-19 (n = 16) infected with lineage B.1.1.7, B.1.351, B.1.1.28.2, and B1 were used to perform PRNT50 against B.1.617.1 variant, and the results were compared with the vaccine recipients' sera samples. The GMT value for vaccine recipients was 88.48 (95% CI: 62.02–126.2) and for recovered cases was 86.85 (95% CI: 52.04–144.9). The sera of BNT162b2 vaccinees, which effectively neutralized B.1.1.7 and P.1 variants, was reduced with the B.1.351 variant [9]. The B.1.617.1 variant performance with vaccine sera was better than that of recovered cases. The results of B.1.1.7 variant neutralization with BBV152 vaccine sera and findings of B.1.617.1 emphasize that this vaccine is robust against emerging mutation and maintains the efficacy of the vaccine (Figure 1E) [10]. Assessment of the clinical efficacy of BBV152 against such variants is underway.

Notes

Author contributions. P. D. Y., G. N. S., and P. A. contributed to the study design, data collection, data analysis, interpretation, and writing and critical review. R. E., R. R. S., A. M. S., D. Y. P., D. A. N., and G. D. contributed to data collection, interpretation, writing and critical review. N. G., S. P., V. K. M., and B. B. contributed to the critical review and finalization of the paper.

Acknowledgments. The authors acknowledge the staff of Indian Council of Medical Research–National Institute of Virology (ICMR-NIV), Pune, including Mr Prasad Sarkale, Mr Shrikant Baradkar, Dr Abhinendra Kumar, Mrs Triparna Majumdar, Mrs Savita Patil, Mr Rajen Lakra, Mr Vishwajeet Dhanure, Ms Pranita Gawande, Mrs Kaumudi Kalele, Mrs Ashwini Waghmare, Ms Jyoti Yemul, Ms Manisha Dudhmal, Ms Aasha Salunkhe, and Mr Chetan Patil for extending excellent technical support. The study was approved by the Institutional Biosafety Committee and Institutional Human Ethics Committee of ICMR-NIV, Pune, India, under the project "Molecular epidemiological analysis of SARS-COV-2 circulating in different regions of India" (20-3-18N).

Financial support. This work was supported by the Indian Council of Medical Research (ICMR), New Delhi, at ICMR–National Institute of Virology, Pune, under intramural funding "COVID-19."

Potential conflicts of interest. The authors: No reported conflicts of interest. All authors have submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Conflicts that the editors consider relevant to the content of the manuscript have been disclosed.

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Clinical Infectious Diseases® **2022;74(2):366–8**
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DOI: 10.1093/cid/ciab411

Is 3 Feet of Physical Distancing Enough?

To the Editor—van den Berg et al [1] conducted a study of students and staff to compare rates of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) transmission between those exposed to a 3-feet versus 6-feet physical-distancing policy in classrooms. Their study has catalyzed a rich scientific discussion about physical distancing measures in schools [2]. While the authors correctly indicated that there were no statistical differences between their comparisons, we argue that the point estimate and 95% confidence interval (CI) for a subset of their results do not wholly support their conclusion

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